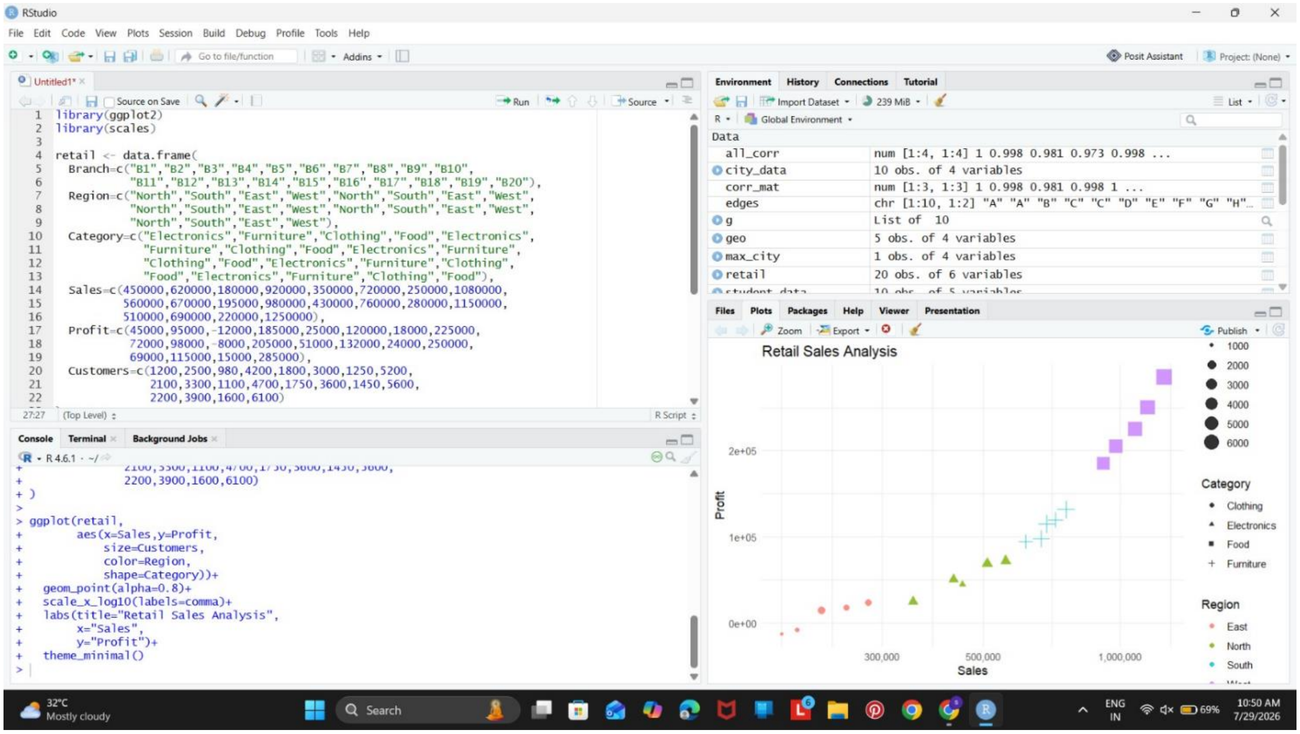
CO1-AT3-Visualization Design Exercise

1.

A bubble chart is selected because it can represent multiple variables in a single visualization. Sales is displayed on the X-axis and Profit on the Y-axis, while the size of each bubble represents the number of customers. Different colours are used to distinguish regions, and different shapes identify product categories, making comparisons easier across branches. Since the sales values vary significantly, a logarithmic scale is applied to the Sales axis to reduce skewness and improve readability. This visualization enables the identification of relationships between sales, profit, customer count, region, and product category in an effective and visually clear manner.

```
library(ggplot2)

library(scales)


retail <- data.frame(

Branch=c("B1","B2","B3","B4","B5","B6","B7","B8","B9","B10",

"B11","B12","B13","B14","B15","B16","B17","B18","B19","B20"),

Region=c("North","South","East","West","North","South","East","West",

"North","South","East","West","North","South","East","West",

"North","South","East","West"),

Category=c("Electronics","Furniture","Clothing","Food","Electronics",

"Furniture","Clothing","Food","Electronics","Furniture",

"Clothing","Food","Electronics","Furniture","Clothing",

"Food","Electronics","Furniture","Clothing","Food"),

Sales=c(450000,620000,180000,920000,350000,720000,250000,1080000,
```

```r
560000,670000,195000,980000,430000,760000,280000,1150000,
510000,690000,220000,1250000),
Profit=c(45000,95000,-12000,185000,25000,120000,18000,225000,
72000,98000,-8000,205000,51000,132000,24000,250000,
69000,115000,15000,285000),
Customers=c(1200,2500,980,4200,1800,3000,1250,5200,
2100,3300,1100,4700,1750,3600,1450,5600,
2200,3900,1600,6100)
)

ggplot(retail,
aes(x=Sales,y=Profit,
size=Customers,
color=Region,
shape=Category))+
geom_point(alpha=0.8)+
scale_x_log10(labels=comma)+
labs(title="Retail Sales Analysis",
x="Sales",
y="Profit")+
theme_minimal()
```



2.

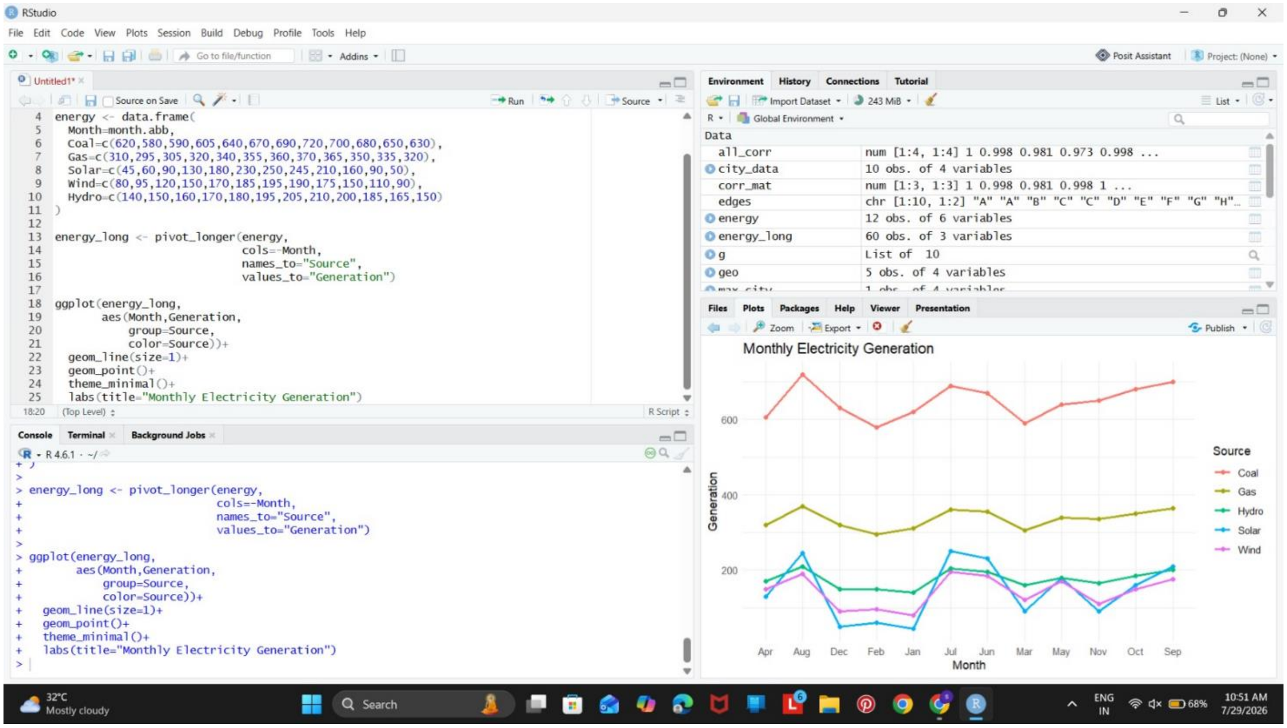
For the Cartesian coordinate system, a line chart is used because it effectively displays the monthly trend of electricity generation for different energy sources. The X-axis represents the months, while the Y-axis represents the electricity generated, and each energy source is shown using a different coloured line. This makes it easy to compare variations and identify trends over time. For the Polar coordinate system, a stacked bar chart is transformed into a circular layout using polar coordinates, which highlights seasonal or cyclical patterns in electricity generation. While the polar chart is visually attractive and useful for presenting cyclic data, the Cartesian line chart provides better analytical insight because it allows more accurate comparison of values and trends.

Cartesian Coordinates

library(ggplot2)

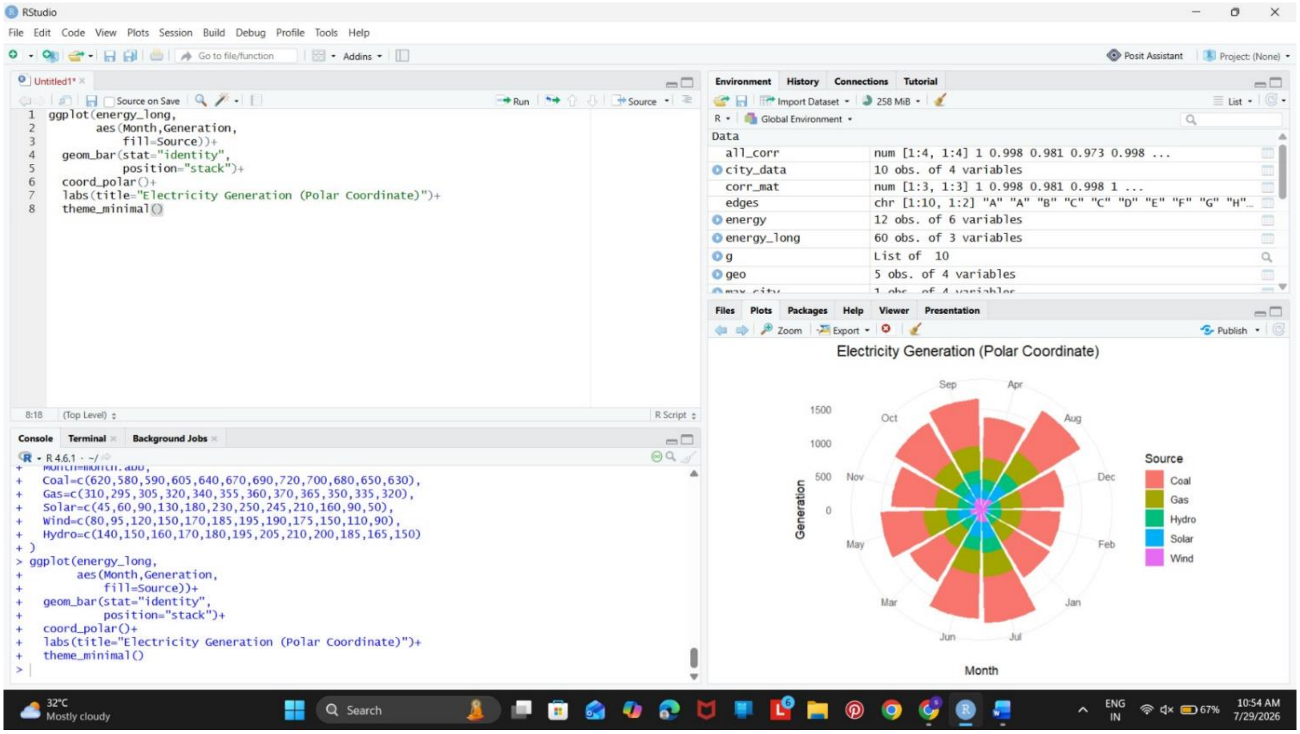
library(tidyr)

```r
energy <- data.frame(
Month=month.abb,
Coal=c(620,580,590,605,640,670,690,720,700,680,650,630),
Gas=c(310,295,305,320,340,355,360,370,365,350,335,320),
Solar=c(45,60,90,130,180,230,250,245,210,160,90,50),
Wind=c(80,95,120,150,170,185,195,190,175,150,110,90),
Hydro=c(140,150,160,170,180,195,205,210,200,185,165,150)
)

energy_long <- pivot_longer(energy,
cols=-Month,
names_to="Source",
values_to="Generation")

ggplot(energy_long,
aes(Month,Generation,
group=Source,
color=Source))+
geom_line(size=1)+
geom_point()+
theme_minimal()+
labs(title="Monthly Electricity Generation")
```



Polar Coordinates

```
ggplot(energy_long,
aes(Month,Generation,
fill=Source))+
geom_bar(stat="identity",
position="stack")+
coord_polar()+
labs(title="Electricity Generation (Polar Coordinate)")+
theme_minimal()
```

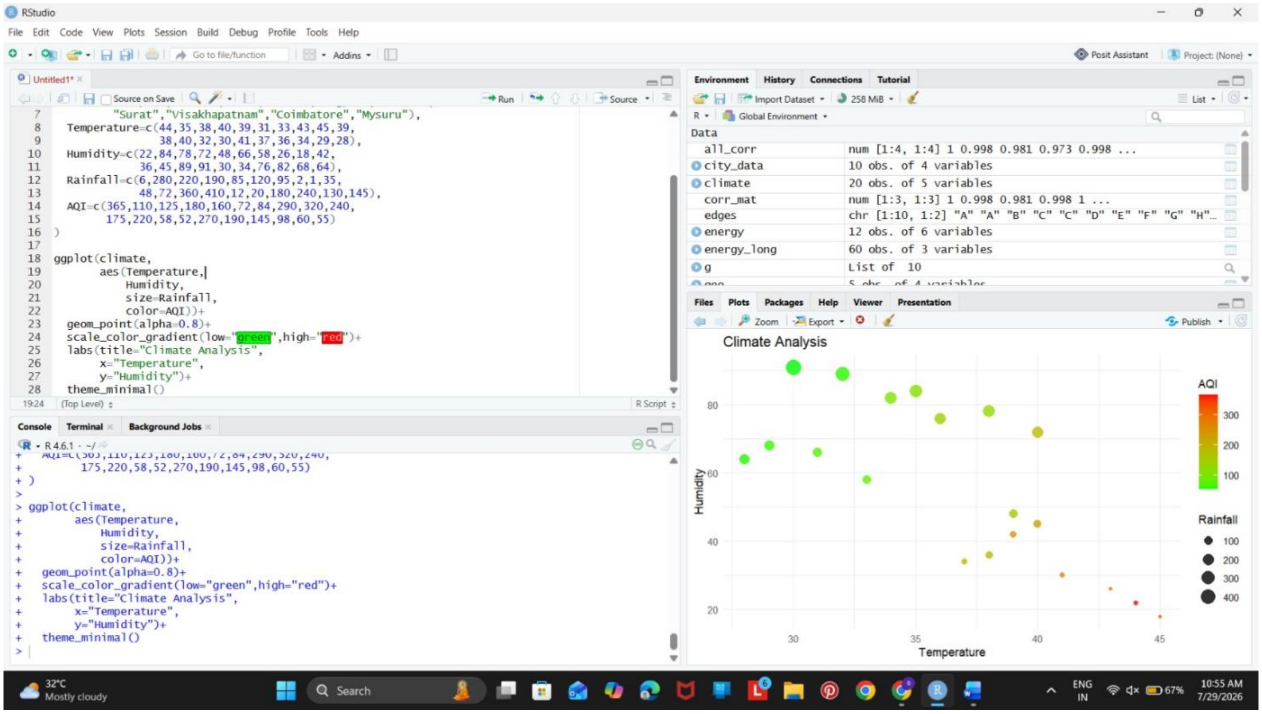


3.

A heatmap is chosen to represent the climate dataset because it effectively displays multiple environmental variables using colour intensity. The cities are represented along one axis, while the climate parameters such as Temperature, Humidity, Rainfall, and AQI are displayed on the other axis. A sequential colour palette is used so that darker colours indicate higher values and lighter colours indicate lower values, making it easy to identify patterns, extreme values, and variations across cities. This visualization improves interpretation, enhances accessibility, and allows quick comparison of all climate parameters within a single graph.

```
library(ggplot2)

climate <- data.frame(

City=c("Delhi","Mumbai","Chennai","Kolkata","Hyderabad",

"Bangalore","Pune","Ahmedabad","Jaipur","Lucknow",
```

```
"Bhopal","Patna","Goa","Kochi","Nagpur","Indore",
"Surat","Visakhapatnam","Coimbatore","Mysuru"),
Temperature=c(44,35,38,40,39,31,33,43,45,39,
38,40,32,30,41,37,36,34,29,28),
Humidity=c(22,84,78,72,48,66,58,26,18,42,
36,45,89,91,30,34,76,82,68,64),
Rainfall=c(6,280,220,190,85,120,95,2,1,35,
48,72,360,410,12,20,180,240,130,145),
AQI=c(365,110,125,180,160,72,84,290,320,240,
175,220,58,52,270,190,145,98,60,55)
)

ggplot(climate,
aes(Temperature,
Humidity,
size=Rainfall,
color=AQI))+
geom_point(alpha=0.8)+
scale_color_gradient(low="green",high="red")+
labs(title="Climate Analysis",
x="Temperature",
y="Humidity")+
theme_minimal()
```



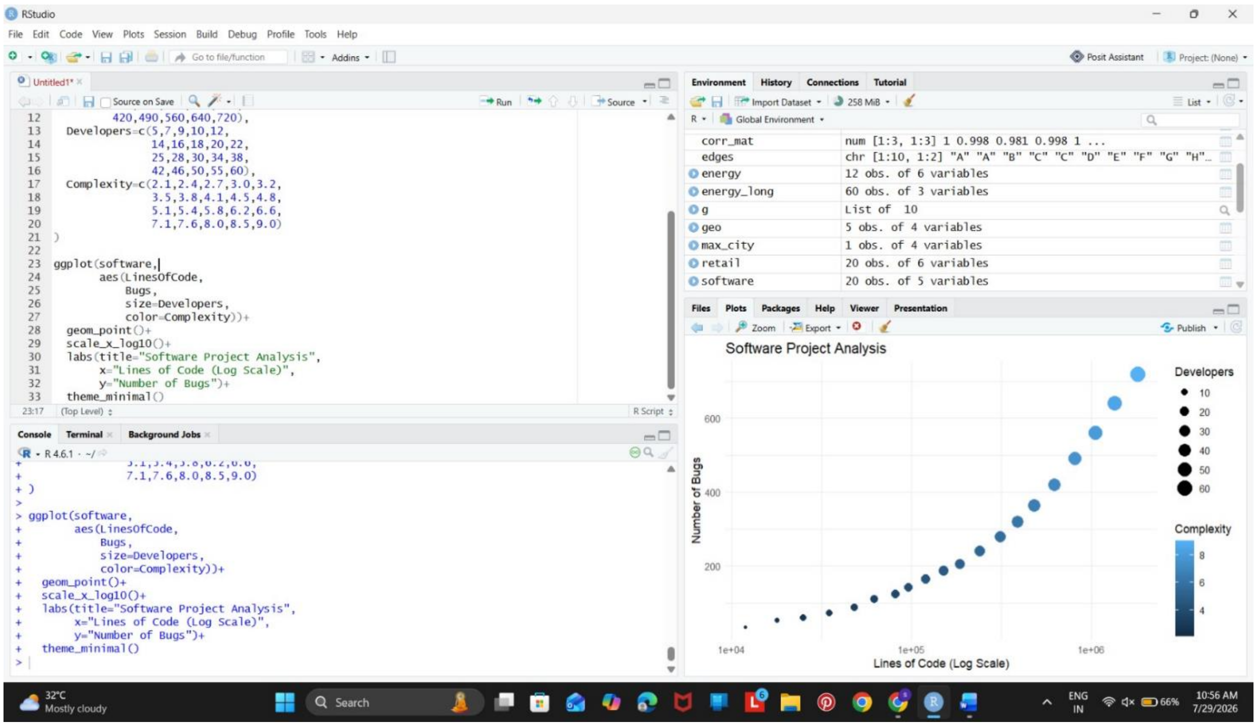
4.

A chart with a logarithmic axis transformation is selected because the Lines of Code values are highly skewed, ranging from a few thousand to nearly two million. Applying a logarithmic scale compresses the larger values while preserving the relative differences between observations, making both small and large projects visible within the same graph. This transformation improves readability, prevents large values from dominating the visualization, and enables a more meaningful comparison of software projects based on their code size, bugs, developers, and complexity.

```
library(ggplot2)

software <- data.frame(

Team=paste("T",1:20,sep=""),

LinesOfCode=c(12000,18000,25000,35000,48000,
```

```
62000,81000,96000,120000,150000,
185000,240000,310000,390000,480000,
620000,810000,1050000,1350000,1800000),
Bugs=c(35,52,60,72,88,
110,125,142,165,188,
205,240,280,320,365,
420,490,560,640,720),
Developers=c(5,7,9,10,12,
14,16,18,20,22,
25,28,30,34,38,
42,46,50,55,60),
Complexity=c(2.1,2.4,2.7,3.0,3.2,
3.5,3.8,4.1,4.5,4.8,
5.1,5.4,5.8,6.2,6.6,
7.1,7.6,8.0,8.5,9.0)
)

ggplot(software,
aes(LinesOfCode,
Bugs,
size=Developers,
color=Complexity))+
geom_point()+
scale_x_log10()+
```

```
labs(title="Software Project Analysis",
x="Lines of Code (Log Scale)",
y="Number of Bugs")+
theme_minimal()
```

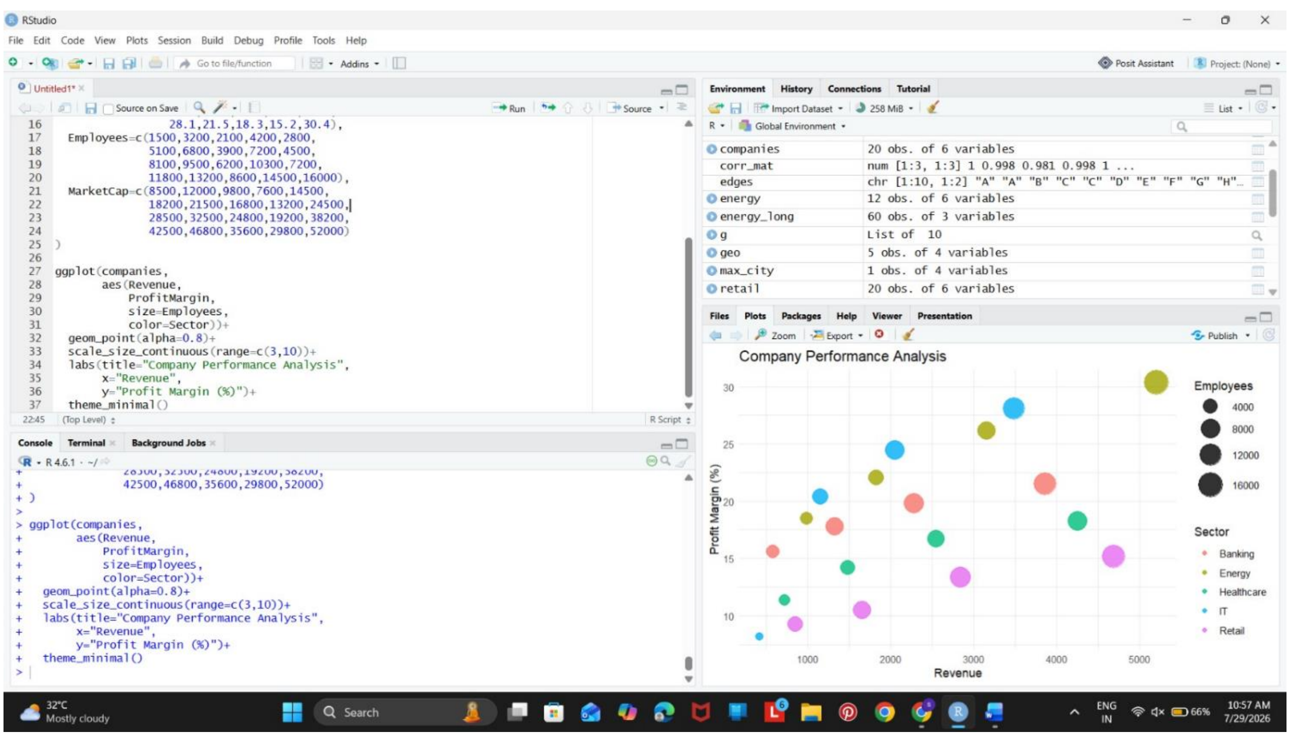


5.

A bubble chart is used to visualize the company performance data because it can represent multiple variables simultaneously. Revenue is displayed on the X-axis and Profit Margin on the Y-axis, while the size of each bubble represents the number of employees. Different colours are assigned to distinguish industry sectors, making it easier to compare companies from different domains. This integrated visualization effectively shows the relationship between revenue, profitability, workforce strength, and industry sector, while reducing visual clutter and improving overall interpretability.

```
library(ggplot2)
companies <- data.frame(
Company=paste("C",1:20,sep=""),
Sector=c("IT","Banking","Healthcare","Retail","Energy",
"IT","Banking","Healthcare","Retail","Energy",
"IT","Banking","Healthcare","Retail","Energy",
"IT","Banking","Healthcare","Retail","Energy"),
Revenue=c(420,580,720,850,980,
1150,1320,1480,1650,1820,
2050,2280,2550,2840,3150,
3480,3860,4250,4680,5200),
ProfitMargin=c(8.2,15.6,11.4,9.3,18.5,
20.4,17.8,14.2,10.5,22.1,
24.5,19.8,16.7,13.4,26.2,
28.1,21.5,18.3,15.2,30.4),
Employees=c(1500,3200,2100,4200,2800,
5100,6800,3900,7200,4500,
8100,9500,6200,10300,7200,
11800,13200,8600,14500,16000),
MarketCap=c(8500,12000,9800,7600,14500,
18200,21500,16800,13200,24500,
28500,32500,24800,19200,38200,
42500,46800,35600,29800,52000)
)
```

```
ggplot(companies,
aes(Revenue,
ProfitMargin,
size=Employees,
color=Sector))+
geom_point(alpha=0.8)+
scale_size_continuous(range=c(3,10))+
labs(title="Company Performance Analysis",
x="Revenue",
y="Profit Margin (%)")+
theme_minimal()
```



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